

GATE BT

Biotechnology

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 18 questions (28%)

Moderate: 38 questions (58%)

Difficult: 9 questions (14%)

GENERAL INSTRUCTIONS

GATE BT — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [-0.33]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q2.

1M [-0.33]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q3.

1M [-0.33]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q4.

1M [-0.33]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q5.

1M [-0.33]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q6.

1M [-0.33]

[Microbiology]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q7.

1M [no -ve]

[Bioprocess Engineering]

Batch process is:

*Enter your answer here***Q8.**

1M [-0.33]

[Immunology]

Antibody produced by:

- (A) T cells
- (B) B cells
- (C) Macrophages
- (D) None

Q9.

1M [-0.33]

[Bioinformatics]

BLAST is for:

- (A) Sequence alignment
- (B) Structure prediction
- (C) Phylogeny
- (D) None

Q10.

1M [-0.33]

[Cell Biology]

Chromosome number in daughter:

- (A) Halved
- (B) Same
- (C) Doubled
- (D) Variable

Q11.

1M [-0.33]

[Bioprocess Engineering]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q12.

1M [-0.33]

[Bioprocess Engineering]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q13.

1M [-0.33]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q14.

1M [-0.33]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q15.

1M [-0.33]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q16.

1M [-0.33]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q17.

1M [-0.33]

[Bioprocess Engineering]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q18.

1M [-0.33]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q19.

1M [no -ve]

[Immunology]

Antibody produced by:

*Enter your answer here***Q20.**

1M [-0.33]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q21.

1M [-0.33]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q22.

1M [-0.33]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q23.

1M [-0.33]

[Microbiology]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q24.

1M [no -ve]

[Bioprocess Engineering]

Batch process is:

*Enter your answer here***Q25.**

1M [-0.33]

[Bioinformatics]

BLAST is for:

- (A) Sequence alignment
- (B) Structure prediction
- (C) Phylogeny
- (D) None

Q26.

1M [-0.33]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q27.

1M [no -ve]

[Cell Biology]

Chromosome number in daughter:

*Enter your answer here***Q28.**

1M [no -ve]

[Bioprocess Engineering]

Batch process is:

*Enter your answer here***Q29.**

1M [-0.33]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q30.

1M [-0.33]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q31.

2M [-0.66]

[Immunology]

Antibody produced by:

- (A) T cells
- (B) B cells
- (C) Macrophages
- (D) None

Q32.

2M [-0.66]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q33.

2M [-0.66]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q34.

2M [-0.66]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q35.

2M [-0.66]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q36.

2M [-0.66]

[Microbiology]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q37.

2M [-0.66]

[Bioprocess Engineering]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q38.

2M [-0.66]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q39.

2M [-0.66]

[Cell Biology]

Chromosome number in daughter:

- (A) Halved
- (B) Same
- (C) Doubled
- (D) Variable

Q40.

2M [-0.66]

[Cell Biology]

Chromosome number in daughter:

- (A) Halved
- (B) Same
- (C) Doubled
- (D) Variable

Q41.

2M [-0.66]

[Genetics]

DNA stands for:

- (A) Deoxyribonucleic Acid
- (B) Ribonucleic
- (C) Dinitrogen Acid
- (D) None

Q42.

2M [-0.66]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q43.

2M [-0.66]

[Microbiology]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q44.

2M [no -ve]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q45.

2M [-0.66]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q46.

2M [no -ve]

[General Aptitude]

Choose similar to 'catalyst':

*Enter your answer here***Q47.**

2M [no -ve]

[Engineering Mathematics]

Normal distribution mean = median = ?

*Enter your answer here***Q48.**

2M [-0.66]

[Bioinformatics]

BLAST is for:

- (A) Sequence alignment
- (B) Structure prediction
- (C) Phylogeny
- (D) None

Q49.

2M [no -ve]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q50.

2M [-0.66]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q51.

2M [no -ve]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q52.

2M [-0.66]

[Bioinformatics]

BLAST is for:

- (A) Sequence alignment
- (B) Structure prediction
- (C) Phylogeny
- (D) None

Q53.

2M [-0.66]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q54.

2M [no -ve]

[Cell Biology]

Chromosome number in daughter:

*Enter your answer here***Q55.**

2M [-0.66]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q56.

2M [-0.66]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q57.

2M [-0.66]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q58.

2M [-0.66]

[Bioprocess Engineering]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q59.

2M [-0.66]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q60.

2M [-0.66]

[Engineering Mathematics]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q61.

2M [-0.66]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q62.

2M [no -ve]

[Bioinformatics]

BLAST is for:

- (A) Sequence alignment
- (B) Structure prediction
- (C) Phylogeny
- (D) None

Q63.

2M [-0.66]

[Biochemistry]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q64.

2M [-0.66]

[General Aptitude]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Q65.

2M [-0.66]

[Genetics]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA